

Route-F P54: Source-Preserving One-Loop Matching and the Upper Heavy-Vector Yukawa/Kinetic Contribution

P54PQ-v2 four-dimensional mainline

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Abstract

We derive the full first-order transport law for the kernel, interaction source and induced contact response (K, J, C). We calculate a previously missing upper-PS heavy-vector contribution using the actual Spin(10) generators and cached same-action stationary Hessian. Dimensional numerator terms supply finite constants which must not be dropped before minimal subtraction. Scalar kinetic matching generates conjugate bidoublet Yukawa coefficients as well as the original PS tensors. Different-colour diquark probes reveal upper scalar-exchange contact terms that the earlier same-colour probes could not see. The one-loop algebra is complete; the diagram content is explicitly a subset. The complete finite matching, fermion-inclusive physical light state and constrained flavor fit are *not* closed by this work.

1 Fixed prescription and exact scope

The action and field content remain P54PQ-v2: $3\,16_F + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}$. The only fundamental family spurions are the two complex symmetric matrices $h_{\text{raw}}, f_{\text{raw}}$. No extra spatial dimension, geometry-dependent particle mass, lattice scan, or new fit parameter is introduced.

This calculation uses dimensional regularization (DR, not dimensional *reduction*), $d = 4 - 2\epsilon$, $\overline{\text{MS}}$ subtraction, background-field Landau gauge, and the existing fixed-VEV tadpole prescription. Open fermion lines need no anomalous γ_5 trace here. The upper stationary PS background, in the previous reference units, is

$$(\omega_U, \sigma_U, v_{s,U}) = (1.00081030592151, 0, 0.254277697083708). \quad (1)$$

It is not the lower broken background $(1, 0.1265, 0.25)$. Its active scalar parents need not have positive curvature; they are retained fields of a Wilson matching calculation, not integrated Gaussian instabilities. Only the 55 positive physical upper scalars are integrated at the upper scalar step. There are 24 upper eaten directions, 248 real active parent coordinates, and one physical PQ direction.

We read the existing content-addressed Hessian rather than recompute it. The historical gauge benchmark gives

$$g_U = 0.562602889985323, \quad \mu_U = 0.563058770438537, \quad M_V^2 = 0.317035178967757. \quad (2)$$

All 24 coset vectors share this mass. These numbers do not recertify the historical dimensionful unification scales or phenomenological viability.

The computed graphs are the heavy-vector correction to the active Yukawa vertex, the vector contribution to the fermion kinetic metric, and the vector-scalar contribution to the active scalar kinetic metric. Eliminated scalar masses in the last graph are kept exactly. Retained scalar masses are expanded to leading order in the hard region. Their largest $|m^2|/M_V^2$ is 0.328605, so an accurate numerical fit requires a controlled remainder or higher matching orders in this expansion. We do not substitute the previously computed *broken-background* scalar-cubic metric into this upper-PS package.

2 Why the source-dependent effective action is required

Let I denote a declared collection of external current slots. Choose the Euclidean Gaussian sign convention

$$S_2[\phi, I] = \frac{1}{2}\phi^T K \phi - \phi^T J I - \frac{1}{2}I^T C I. \quad (3)$$

Then integrating ϕ gives the response

$$F = C + J^T K^{-1} J, \quad S_{\text{ind}} = -\frac{1}{2}I^T F I. \quad (4)$$

The transpose is intentional: ϕ, I are real-field/current coordinates analytically continued in momentum. It is not a Hermitian conjugate at complex momentum. Matrix spectral positivity concerns the appropriate physical real-energy representation, not arbitrary complex K .

2.1 Exact elimination and its first derivative

Partition retained and eliminated coordinates as

$$K = \begin{pmatrix} A & B \\ B^T & D \end{pmatrix}, \quad J = \begin{pmatrix} J_r \\ J_h \end{pmatrix}, \quad R = D^{-1}. \quad (5)$$

Completing the square gives

$$K'_r = A - B R B^T, \quad (6)$$

$$J'_r = J_r - B R J_h, \quad (7)$$

$$C' = C + J_h^T R J_h. \quad (8)$$

Write every input as $X = X_0 + \ell \delta X + O(\ell^2)$, with a formal loop-counting parameter ℓ . The displayed numerical δX already contains its $1/(16\pi^2)$ coefficient. Since $\delta R = -R\delta D R$, the complete linearized map is

$$\delta K'_r = \delta A - \delta B R B^T - B R \delta B^T + B R \delta D R B^T, \quad (9)$$

$$\delta J'_r = \delta J_r - \delta B R J_h - B R \delta J_h + B R \delta D R J_h, \quad (10)$$

$$\delta C' = \delta C + \delta J_h^T R J_h + J_h^T R \delta J_h - J_h^T R \delta D R J_h. \quad (11)$$

All unvaried quantities here are at tree order. The response derivative is

$$\delta F = \delta C + \delta J^T K_0^{-1} J_0 + J_0^T K_0^{-1} \delta J - J_0^T K_0^{-1} \delta K K_0^{-1} J_0. \quad (12)$$

Proposition 1 (Order-independent one-loop transport). *If all eliminated blocks along two nested elimination paths are invertible, and both paths start from the same source-dependent action and expansion, their final (K, J, C) and their first variations agree.*

Proof. The tree maps are the same completion of the square in the joint eliminated variables. The resulting rational matrix functions agree on their common invertibility domain. Differentiating that identity gives Eqs. (9)–(11); the chain rule proves the statement. No assumption of commuting mass blocks is used. $\square$

This is an algebraic theorem, not a claim that separately truncated local EFTs automatically have identical Wilson coefficients. Local expansion, IR subtraction, operator reduction, and scale evolution must also be performed consistently. Singular blocks cannot be repaired by an unannounced pseudoinverse.

2.2 Where the missing finite diagrams enter

For classical heavy solution $\Phi_h^{\text{cl}}[\varphi, I]$, the formal one-loop matching functional has the structure

$$\Delta\Gamma^{(1)}[\varphi, I] = \left[\frac{1}{2} \text{STr} \log \mathcal{M}_{\text{UV}}[\varphi, \Phi_h^{\text{cl}}, I] - \frac{1}{2} \text{STr} \log \mathcal{M}_{\text{EFT},0}[\varphi, I] + \Gamma_{\text{CT}} \right]_{\text{local,ren}}. \quad (13)$$

Here φ denotes retained fields, not loop counting; $\mathcal{M}$ is the full gauge-fixed fluctuation operator. The supertrace includes the correct real-boson, fermion and ghost multiplicities. The EFT subtraction retains soft diagrams and tree Wilson insertions generated by heavy elimination. The hard-region formulation of this subtraction is discussed in Ref. [1]. In the convention of Eq. (3),

$$\delta K = \Delta\Gamma_{\varphi\varphi}^{(2)}, \quad \delta J = -\Delta\Gamma_{\varphi I}^{(2)}, \quad \delta C = -\Delta\Gamma_{II}^{(2)}. \quad (14)$$

For example, the logarithm obeys

$$\partial_x \partial_y \frac{1}{2} \text{STr} \log \mathcal{M} = \frac{1}{2} \text{STr} (\mathcal{M}^{-1} \mathcal{M}_{xy} - \mathcal{M}^{-1} \mathcal{M}_x \mathcal{M}^{-1} \mathcal{M}_y). \quad (15)$$

Thus retaining only a mass Hessian is insufficient: derivatives of the source-dependent fluctuation operator and its EFT counterpart are essential. A one-loop shift of the stationary eliminated field does not add a new first-order term to the stationary tree functional, but the declared vacuum counterterms and tadpole prescription must still be used.

Our finite set of scalar-exchange current probes is not a complete local operator basis. Full matching additionally needs appropriate Lorentz, gauge and family structures, direct vector exchange, box contributions, composite-current renormalization, and operator mixing. A vanishing C in a selected probe sector does not establish their absence.

3 Actual upper heavy-vector finite kernels

3.1 Conventions and dimensionally finite terms

Let R_A be real antisymmetric scalar generators and T_A Hermitian fermion generators, with $T_A = iR_{F,A}$. The original Weyl convention is

$$\mathcal{L}_Y = -\frac{1}{2} \psi^T Y^a \psi s_a + \text{h.c.}, \quad (Y^a)^T = Y^a, \quad Z_\psi = 1 + k_\psi, \quad Z_s = 1 + k_s. \quad (16)$$

Write $L_0 = 16\pi^2$ and

$$B_0(0; x, y) = \Delta_\epsilon + b(x, y; \mu), \quad (17)$$

$$\begin{aligned} b(x, y; \mu) &= - \int_0^1 \log \frac{(1-t)x + ty}{\mu^2} dt \\ &= 1 - \frac{x \log(x/\mu^2) - y \log(y/\mu^2)}{x - y}. \end{aligned} \quad (18)$$

In particular $b(x, 0) = 1 - \log(x/\mu^2)$ and $b(x, x) = -\log(x/\mu^2)$. The integral form controls both degenerate and one-zero limits.

The heavy Landau vector propagator is transverse. At zero external momentum the fermion triangle gives $d-1$, whereas the scalar momentum coefficient is $4(d-1)/d$. Therefore

$$[(d-1)B_0]_{\overline{\text{MS}}} = 3b - 2, \quad (19)$$

$$\left[\frac{4(d-1)}{d} B_0 \right]_{\overline{\text{MS}}} = 3b - \frac{1}{2}. \quad (20)$$

These constants would be missed by setting $d=4$ before multiplying the UV pole. They are not adjustable matching nuisances.

For the massless fermion self-energy the numerator is

$$\begin{aligned} & \gamma^\mu (\not{p} - \not{k}) \gamma^\nu \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) \\ &= (3-d) \not{p} + (d-1) \not{k} - 2 \frac{k \cdot p}{k^2} \not{k}. \end{aligned} \quad (21)$$

Expand the massless denominator to first order in p and average $k_\mu k_\nu \mapsto g_{\mu\nu} k^2/d$. The coefficient of $\not{p}$ becomes

$$a(d) = 5 - d - \frac{4}{d} = \frac{3}{2}\epsilon + O(\epsilon^2). \quad (22)$$

Since the inverse propagator is $\not{p} - \Sigma$, its finite kinetic coefficient in Landau gauge is

$$k_\psi^V = -\frac{3g_U^2}{2L_0} \sum_A T_A^2. \quad (23)$$

The absence of a Landau logarithm in this particular kinetic graph does not imply that the finite contribution vanishes.

3.2 Independent Abelian mass check

A neutral scalar mass insertion for a charged Dirac fermion provides a check independent of the P54 group sums. Factoring out $g^2 q^2/L_0$ and putting $l = \log(M^2/\mu^2)$, the mass and slash self-energies in R_ξ give

$$\Sigma_m/m = 1 - 3l + u_\xi, \quad \Sigma_\not{p} = \frac{3}{2} - u_\xi, \quad u_\xi = \xi(1 - l - \log \xi), \quad u_0 = 0. \quad (24)$$

Thus canonical mass matching is gauge independent in this limit:

$$\frac{\delta m}{m} = \frac{g^2 q^2}{L_0} \left(\frac{5}{2} - 3l \right). \quad (25)$$

The same constant/logarithm structure occurs in the neutral-vector part of the light-fermion matching formula in Ref. [2]. That comparison does not import its Standard Model charges, vacuum convention, or full threshold into P54. The executable tests also check the d -dependent Laurent limits directly. RG logarithms alone cannot verify these finite constants.

3.3 Vertex and full active scalar metric for this subset

Let A be the 328×248 active embedding, H the orthonormal 328×55 upper heavy eigenbasis, and m_j^2 its positive spectrum. The finite active vertex is

$$\delta V^a = -\frac{g_U^2}{L_0} \sum_A T_A^T Y^a T_A [3b(M_V^2, 0; \mu) - 2]. \quad (26)$$

To avoid confusing the active embedding with a gauge index, set $v_{A,a} = R_A A_a$ and define

$$w_0 = 3b(M_V^2, 0; \mu) - \frac{1}{2}, \quad (27)$$

$$W = w_0(1 - HH^T) + H \text{diag}_j[3b(M_V^2, m_j^2; \mu) - \frac{1}{2}]H^T. \quad (28)$$

Then the complete vector-scalar state sum for the active metric, at the stated retained-mass expansion order, is

$$(k_s^V)_{ab} = -\frac{g_U^2}{L_0} \sum_A v_{A,a}^T W v_{A,b}. \quad (29)$$

The complement is included, not deleted: it contains all retained and zero-mode scalar directions, expanded at zero mass in the hard integral.

At this upper background, eaten directions lie in 54_H , while active scalars lie in $10_H + 126_H$. Consequently their linear $VV s_a$ mass vertices vanish, as do their transitions to eaten scalars through R_A . Goldstone Yukawa tensors vanish because 54_H has no Yukawa coupling. Mixed scalar–fermion–vector triangles at zero external momentum have a scalar vertex proportional to vector momentum and vanish against the Landau projector. Ghosts do not couple to fermions; active linear ghost mass vertices vanish here. Seagulls can affect the *potential*, but not the p^2 metric coefficient calculated in Eq. (29). None of these statements removes the separate scalar-cubic graphs or vacuum counterterm matching.

4 Canonical matching, Ward checks, and generated tensors

For real scalar coordinates the first-order canonical Yukawa tensor is

$$Y_c^a = Y^a + \delta V^a - \frac{1}{2} \left[k_\psi^T Y^a + Y^a k_\psi + \sum_b (k_s)_{ba} Y^b \right]. \quad (30)$$

This external-leg structure is also obtained from finite field maps in Ref. [3]; its $SU(5)$ group factors are not used here. The real metric is essential: a complex scalar metric can include both ordinary and conjugate-bidoublet structures after PS breaking.

The actual Clifford and scalar representations give

$$C_F^{\text{coset}} = 3I_{16}, \quad C_s^{\text{coset}}|_{H,F,L,R} = (3, 7, 6, 6). \quad (31)$$

Squaring the Yukawa intertwiner identity gives

$$\sum_b (C_s)_{ab} Y^b = C_F^T Y^a + Y^a C_F + 2 \sum_A T_A^T Y^a T_A. \quad (32)$$

Since $db/d \log \mu = 2$, Eqs. (23)–(30) imply the independent scale check

$$\frac{d\delta Y_c^a}{d \log \mu} = \frac{3g_U^2}{L_0} (C_F^T Y^a + Y^a C_F) = \beta_{\text{PS}}^a - \beta_{\text{Spin}(10)}^a \Big|_{\text{removed vectors}}. \quad (33)$$

In the auxiliary limit in which every scalar in the vector bubble is treated at zero mass, a parent of coset Casimir c_s gives

$$\delta Y_c = \frac{g_U^2}{L_0} \left[\frac{15}{2} + \frac{3}{4} c_s - 9 \log(M_V^2/\mu^2) \right] Y. \quad (34)$$

Actual hard scalar mass splittings are retained instead of imposing this auxiliary formula on the numerical data.

At the benchmark we obtain

$$k_\psi^V = -0.00901979574931 I_{16}, \quad (35)$$

$$\text{spec}(k_s^V) \subset [-0.0300659858310, -0.000821449346042]. \quad (36)$$

Both corrected kinetic metrics $I + k$ remain positive for this subset. The canonical finite coefficients, in the *previously fixed* six-invariant PS basis, are

UV spurion	PS parent	Direct increment	Mirror increment
h_{raw}	H	0.01771675106	$+0.0001280086893 i$
f_{raw}	F	0.08885260368	$-0.0003536675766 i$
f_{raw}	L	0.06803155989	none
f_{raw}	R	0.06803155989	none

Each entry multiplies its row's UV family matrix. Other direct cross entries vanish for this gauge contribution. Mirror phases are basis conventions, not additional CP parameters. These linear coefficients are distinct from the earlier cubic-spurion pure-Yukawa thresholds. Both live at the same upper PS site, but their sum is still missing the other upper matching pieces. The six-invariant residual is below 4×10^{-15} ; projecting onto only four tensors discards real terms.

5 Numerical source-preserving transport on this background

The physical basis is $(H, F, L, R, \text{PQ}, \text{heavy})$, of dimension $248 + 1 + 55 = 304$. At complex Euclidean $z = p_E^2$ use

$$K_0(z) = B^T H_{\text{scalar}} B + z I_{304}, \quad \delta K(z) = z \text{diag}(k_s^V, 0_{56}). \quad (37)$$

The zero entries specify the *chosen additive graph contribution*, not the unknown complete self-energy. The source increment consists of the computed active vertex and the universal fermion external legs, including on heavy scalar sources:

$$\delta Y_{\text{source}}^A = -\frac{1}{2}(k_\psi^T Y^A + Y^A k_\psi) + \sum_a A_{Aa} \delta V^a. \quad (38)$$

It intentionally excludes scalar external-leg factors, since those are already carried by K . Before scalar elimination $\delta C = 0$ for this additive graph set only. Unknown hard contact graphs remain unknown.

The probe ledger retains the seven P-SPEC1 spinor pairs and adds $u_1^c d_2^c$ and $u_{L,1} d_{L,2}$, where subscripts denote the first two matched colours in the common-phase dictionary. Each pair has h/f and real/imaginary slots, giving 36 real current slots. The old same-colour diquark contractions vanished by antisymmetric colour selection; they could not certify absence of the upper triplet channel. The new action-derived sources yield nonzero upper δC .

We compare direct $304 \rightarrow 129$ reduction with $304 \rightarrow 249 \rightarrow 129$ and the reverse elimination order. The final chart retains whole H, F parents and PQ. Eliminating L, R here is an exact off-shell *algebra test*, not permission to decouple these physically light parents at the upper threshold. At four nonsingular momenta,

z	$\ \delta C_{\text{upper}}\ _F$	error after dropping final δC
0.013	0.2483693	0.9998932
0.27	0.1336376	0.9611380
$-0.11 + 0.07i$	0.4106111	0.6078394
$0.03 + 0.12i$	0.2166991	0.9651812

The error is $\|F_1 - F_2\| / \max(1, \|F_1\|, \|F_2\|)$ for response *increments*, not a cross-section uncertainty. The raw family spurions are factored out. In particular, these large diagnostic differences do not assert large physical radiative corrections.

For an infinitesimal real-field redefinition $\phi = (I + \ell s)\phi_c$, covariance requires

$$\delta K_c = \delta K + s^T K_0 + K_0 s, \quad \delta J_c = \delta J + s^T J_0, \quad \delta C_c = \delta C. \quad (39)$$

For $s = -k_s/2$ the kinetic correction is removed while the curvature becomes $\delta M_c = \delta M - \{k_s, M_0\}/2$. Substitution into Eq. (12) cancels all s terms exactly. Including the scalar leg in J while also retaining the unnormalized K double counts it; this incorrect alternative is detected at all four momenta. Dense synthetic jets additionally exercise nonzero δB and δD , which happen to vanish in portions of the actual sparse sub-contribution. They are labelled algebra tests, not physical inputs.

6 The light-state and fit handoff, after matching closes

In original complex doublet coordinates let

$$D = D_0 + \delta D_B + \delta D_F + \delta D_{\text{match}} - \delta \xi P_{10}, \quad Z = I + k_H. \quad (40)$$

Here δD_F includes the full same-prescription fermionic tadpole counterterm, and all Yukawa matrices are the actual evolved matched ones. Neither unit-spurion tests nor free independent mirror matrices can supply them. The constrained light problem is

$$Dc = 0, \quad c^\dagger Zc = 1, \quad QD_c Q > 0, \quad D_c = Z^{-1/2} D Z^{-1/2}. \quad (41)$$

The kinetic metric alone cannot rotate an exact original-coordinate null ray. It changes its normalization and its canonical representative. For a simple tree zero $D_0 c_0 = 0$, $c_0^\dagger c_0 = 1$, define $Q = 1 - c_0 c_0^\dagger$ and $G = (QD_0 Q)^{-1}$ on $\text{ran } Q$. First-order perturbation theory gives

$$\delta \xi = \frac{c_0^\dagger \delta D_{\text{rest}} c_0}{c_0^\dagger P_{10} c_0}, \quad (42)$$

$$Q \delta c = -G Q (\delta D_{\text{rest}} - \delta \xi P_{10}) c_0, \quad (43)$$

$$\text{Re}(c_0^\dagger \delta c) = -\frac{1}{2} c_0^\dagger k_H c_0. \quad (44)$$

The phase fixes the remaining imaginary parallel part. If the corrected heavy block is not positive or is nearly singular, perturbative light-state promotion is not justified. An exact numerical retuning with one-loop inputs does not create a two-loop calculation.

Up/Dirac-neutrino Yukawas contract with c ; down/charged-lepton vertices use the conjugate-copy dictionary and c^* . All six allowed PS Yukawa tensors must run together, subject to their boundary origin in $h_{\text{raw}}, f_{\text{raw}}$. Subsequent nondegenerate seesaw thresholds, C_{HN} , Weinberg/type-II coefficients and finite kinetic maps must use that same trajectory. The fit objective can then be written

$$\chi^2(\theta, \eta) = [O(\theta, \eta) - O_{\text{data}}]^T V^{-1} [O(\theta, \eta) - O_{\text{data}}] + \chi_{\text{nuisance}}^2(\eta), \quad (45)$$

subject to stationarity, one light doublet, the heavy-gap constraint, perturbative control and the common RG/matching map. Any residual $\eta_Y(\mu)$ must describe a declared uncertainty of this map, not replace uncalculated matrices by independent per-observable parameters. The present work cannot provide the missing map, so no minimization of this physical objective is performed.

7 Reproducibility, closed subgates, and next priority

The executables are

```
route_f/code/verify_p54_upper_vector_matching.py
route_f/code/verify_p54_kjc_finite_matching.py
```

Both accept `--cache-dir /path/to/tmp/p54_full_doublet_cw`. Their JSON/Markdown outputs contain matrices, fixed phases, source hashes, all residuals and the missing-diagram ledger. The vector verifier passes 38/38; the transport verifier passes 108/108. The pre-existing complex canonical matching interface also passes 17/17 in memory.

The contribution composer rejects a different action, gauge, subtraction, tadpole convention, background, scale, momentum, basis, mass expansion or loop order, and rejects duplicated diagram labels. The shared fit consumer now requires full upper/lower (K, J, C) , all-six PS evolution, finite sequential seesaw, full fermionic light feedback and retained-mass error control. Old boolean flags alone no longer open it. These guards are bookkeeping checks, not proofs that a submitted diagram inventory is physically complete.

Priority remains within the fixed action:

1. Complete the upper scalar-cubic kinetic and source-dependent potential/tadpole/Wilson matching in this *same* PS chart, including a controlled retained-mass expansion and a complete operator basis.
2. Construct the lower covariant hard-minus-EFT finite functional and finite sequential seesaw map; do not add the old one-step broken result directly to the upper threshold. One-loop low-energy matching requires common amplitudes and their operator content, as exemplified by Ref. [4].
3. Only then iterate the full complex fermionic light state and the constrained two-spurion fit. Momentum-dependent pole masses and gauge/Nielsen checks remain a later, distinct requirement.

No unrestricted theorem proving the phenomenological theory is claimed.

References

- [1] B. Henning, X. Lu and H. Murayama, *One-loop Matching and Running with Covariant Derivative Expansion*, [arXiv:1604.01019](#).
- [2] S. P. Martin, *Matching relations for decoupling in the Standard Model at two loops and beyond*, Phys. Rev. D 99 (2019) 033007, [arXiv:1812.04100](#), Eq. (2.20).
- [3] *Quantum corrections and the minimal Yukawa sector of SU(5)*, [arXiv:2310.16563v2](#), Appendices A/B; used for external-leg structure, not P54 group factors.
- [4] W. Dekens and P. Stoffer, *Low-energy effective field theory below the electroweak scale: matching at one loop*, [arXiv:1908.05295](#).